Analytics Requirements Specification (ARS)

Technical Architecture, Data Pipelines & Semantic Modeling Specifications

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1. Introduction

This Analytics Requirements Specification (ARS) defines the technical architecture and specification baseline for implementing the E-Commerce Enterprise Data Warehouse inside Microsoft Fabric. It provides low-level requirements for data ingestion, star schema relationships, KPI metric logic, reporting structures, workspace security, and deployment lifecycle governance without relying on hardcoded scripts or raw code.

2. Data Sources

The platform ingests four operational CSV extracts landed in Microsoft Fabric OneLake storage:

Source Extract File	Record Volume	Keys & Relationships	Attributes & Target Field Specifications
`users.csv`	1,000 rows	`user_id` (Primary Key)	User name, email address, account join date, and loyalty tier (`Silver` or `Platinum`).
`products.csv`	50 rows	`product_id` (Primary Key)	Product name, product category (`Electronics`, `Garden & Outdoor`, `Grocery & Wellness`, `Apparel`), base standard price.
`orders.csv`	5,000 rows	`order_id` (PK), `user_id` (FK)	Order purchase timestamp, fulfillment status (`placed`, `shipped`, `delivered`, `cancelled`).
`order_items.csv`	12,000 rows	`order_item_id` (PK), `order_id` (FK), `product_id` (FK)	Purchased unit quantity (1 to 5), checkout unit price paid (inclusive of ±5% random variance from base price).

3. Data Ingestion and Transformation

Data ingestion and processing follow a structured two-stage architecture inside Microsoft Fabric:

Pipeline Stage	Fabric Technology	Transformation & Business Processing Rules
Staging Ingestion	Dataflows Gen2	Extracts CSV files from OneLake and loads raw data into staging tables (`stg` schema) with enforced data typing and null validation.
Dimensional Loading	Fabric Warehouse Batch Engine	Joins staging header and item tables, generates integer surrogate keys, resolves role-playing date dimensions, and derives line totals.
Quality Enforcements	Data Pipeline Validations	Enforces sequence validation ensuring order timestamps occur strictly after user join dates. Violations route to audit tables.

4. Data Model and Storage

The data storage architecture is organized as a Kimball Star Schema within the Microsoft Fabric Data Warehouse (`dw` schema) optimized for high-performance direct querying.

Table Object Name	Model Role	Surrogate Key	Business Natural Keys & Attributes
`dw.dim_user`	Dimension	`user_key` (Integer)	`user_id` (Natural Key), user name, email, loyalty tier, account join date.
`dw.dim_product`	Dimension	`product_key` (Integer)	`product_id` (Natural Key), product name, category name, standard base price.
`dw.dim_date`	Dimension	`date_key` (YYYYMMDD)	Calendar date, calendar year, quarter, month name, day of week, weekend indicator.
`dw.fact_order_line`	Fact Table	`order_line_key` (BigInt)	FKs to user, product, and order date dimensions. Degenerate: `order_id`, `status`. Measures: quantity, unit price, line total.

5. Business Rules and KPIs

Business measures and Key Performance Indicators (KPIs) are modeled centrally in the semantic model to guarantee metric consistency across all downstream reports:

KPI Measure Name	Technical Logic & Mathematical Definition	Display Format
Total Net Revenue	Sum of all line-item extended amounts (quantity multiplied by unit price paid).	Currency (`$#,##0.00`)
Total Orders	Distinct count of unique order identification numbers across transaction lines.	Integer (`#,##0`)
Average Order Value	Total Net Revenue divided by Total Orders.	Currency (`$#,##0.00`)
Total Units Sold	Sum of all item quantities purchased across order lines.	Integer (`#,##0`)
Average Selling Price	Arithmetic average of actual checkout unit prices paid across transactions.	Currency (`$#,##0.00`)
Total Price Variance	Sum of price differential (checkout price minus base catalog price) multiplied by item quantity.	Currency (`$#,##0.00`)
Active Customers	Distinct count of unique user keys present in the transaction fact table.	Integer (`#,##0`)

> **Data Granularity Guarantee**
> The fact table maintains an exact atomic line-item granularity matching `order_items.csv` (12,000 records), eliminating fan-out aggregation errors when filtering across product categories or loyalty tiers.

6. Reports and Dashboards

The reporting front-end consists of a 3-sheet interactive dashboard connected directly to the semantic model via Direct Lake mode:

Dashboard Sheet	Slicers & Filter Controls	Visual Component & Axis Field Bindings
1. Sales Overview	Date Range, Order Status	Dual-Axis Line/Column (Month vs Revenue & Orders), Bar Chart (Status vs Revenue), Column Chart (Day of Week vs Avg Revenue).
2. Product Analytics	Product Category, Loyalty Tier	Bar Chart (Top 10 Products by Revenue), Grouped Column (Category vs Revenue & Units), Comparison Column (Base Price vs Paid Unit Price).
3. User Insights	Loyalty Tier, Category	Donut Chart (Tier vs Revenue Share), Line Chart (Join Cohort vs Spending), Column Chart (Loyalty Tier vs Basket Size).

7. Security and Access

Security governance combines Microsoft Fabric workspace role-based permissions with fine-grained semantic model data filtering:

Security Layer	Access Specification & Governance Policy
Workspace Roles	Admin permissions assigned to Lead Engineers, Member access for BI Developers, and Viewer roles for business stakeholders.
Data Level Security	Dynamic Row-Level Security (RLS) restricts data views based on authenticated user principal names (UPN) for sensitive customer fields.

8. Testing and Deployment

Quality assurance and release management utilize Microsoft Fabric Deployment Pipelines across three isolated stages:

Deployment Stage	Validation Protocol & Quality Gate Benchmark
Development (DEV)	Pipeline verification, surrogate key mapping checks, and semantic model measure logic validation.
Test / UAT Stage	Data reconciliation testing matching 100% of financial totals and row counts between raw CSVs and report visuals.

9. Monitoring and Support

System operational health is managed through continuous automated monitoring and capacity tracking:

Monitoring Domain	Technical Strategy & Incident SLA
Pipeline Execution	Fabric Monitoring Hub tracks pipeline job refreshes and triggers automated failure notification alerts with a 15-minute SLA.
Capacity Utilization	Microsoft Fabric Capacity Metrics app monitors Capacity Unit (CU) consumption, alerting when peak usage exceeds 80% threshold.